

# WHOSE LABEL? ACTOR INTENT VERSUS LISTENER PERCEPTION IN CREMA-D SPEECH EMOTION RECOGNITION

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## ABSTRACT

CREMA-D results are almost always scored against the filename emotion, which records an actor’s instruction rather than what a listener heard. The corpus ships the data needed to check that—68,568 voice-only judgements—and it is seldom used. Listeners recover the intended category for 45.6% of clips; sad reaches 18.2% against a 16.7% floor, because 71.8% of sad-intended clips are heard as neutral. We train eight systems and score identical predictions against both label sets. Swapping the key alone moves a fixed system’s accuracy by up to 29.7 points and reorders any table mixing the conventions ( $\tau = 0.58$ ). The intended label is not merely noisier: a classifier recovers acted sadness at 66.6% where listeners reach 18.2%. It is a learnable target with little to do with what anyone hears, and a CREMA-D number’s value and its verdict both turn on a choice papers rarely state.

**Index Terms**— speech emotion recognition, label uncertainty, perceptual annotation, evaluation protocols, CREMA-D

## 1. INTRODUCTION

CREMA-D [1] is among the most heavily used corpora in speech emotion recognition (SER). Its appeal is obvious: 7,442 clips, 91 actors with a deliberately broad demographic spread, six categories, clean recordings, and a permissive licence. The convention in the literature is to read the label off the filename—1001\_TEO\_HAP\_LO is a happy clip—and to report accuracy against it. The tooling follows suit: the most widely used public loader for the corpus obtains its label by splitting the filename on underscores and never opens the rating files.<sup>1</sup> An indirect confirmation is that CREMA-D results are almost always reported on a near-balanced six-way problem, which is a property of the intended labels alone; the perceptual labels are 56.6% neutral.

That label is a record of the direction an actor received. It says what the speaker was asked to convey, and it is fixed before the microphone is switched on. Whether the resulting audio conveys anything of the sort to a listener is a separate empirical question, and CREMA-D is unusual among acted corpora in having already answered it: the release includes per-clip perceptual ratings collected in three presentation conditions, one of which is voice only. Those ratings sit in `processedResults/` and are, as far as we can tell, almost never used for anything beyond reporting inter-rater agreement in passing.

We looked at them. Restricting attention to the audio condition, the one an SER system is actually being asked to emulate, listeners recover the intended category for 45.6% of clips. Sad is recovered

18.2% of the time against a chance floor of 16.7%. The gap is not a curiosity confined to a handful of poor recordings: it is broad, systematic, and it has a direction, which is that almost everything drifts toward neutral.

This paper makes three claims. First, on CREMA-D audio the standard answer key is one that listeners themselves largely cannot reproduce (Section 3). Second, swapping to the perceptual labels moves a fixed system’s reported accuracy by as much as 29.7 points, and scrambles the ordering of a table whose entries were produced under different conventions, which is what a benchmark table for this corpus actually is (Section 5). Third—and this surprised us—the difference between the two keys is not a difference between signal and noise. A plain classifier recovers acted sadness from the audio 3.7 times more often than listeners do, so the filename label is a perfectly learnable target; it is just not a target that has much to do with what anyone hears. Code, the derived label sets and the full result tables are released with the paper.<sup>2</sup>

## 2. RELATION TO PRIOR WORK

That emotion perception varies across observers is old news in psychology [2, 3] and has been argued forcefully within affective computing as well [4, 5]. The SER community has responded mainly on the modelling side. Emotion profiles [6] replace a single categorical decision with a vector of class evidence; soft-target training [7] fits the rater distribution instead of its argmax; and more recent work models individual annotators explicitly rather than pooling them [8]. Corpora designed around perception (IEMOCAP [9], MSP-IMPROV [10], MSP-Podcast [11]) take the perceptual reading as primitive: the reference label is a consensus over annotators, and challenges built on them score against that consensus by construction [12].

Our contribution is not another soft-label objective. It is an argument about which answer key a widely used benchmark should be scored against, made quantitatively, on a corpus where the question can be settled with data the corpus already distributes. The distinction matters because the two lines of work are easily conflated. Soft-label methods are typically evaluated by showing that a new objective improves agreement with a distribution; here we hold the *models* fixed and vary only the scoring key, which separates the effect of the labelling convention from any effect of the training objective. The closest prior observation is the general remark, made in several of the works above, that consensus labels discard information. What has not been shown, to our knowledge, is that on CREMA-D the convention in use is not merely lossy but is a target listeners miss more often than they hit, and that swapping the key reorders the leaderboard.

<sup>1</sup>The line `filename.split("_")` in the HuggingFace `crema-d` loading script.

<sup>2</sup><https://github.com/24f1002043/icassp>

**Table 1.** Voice-only listener behaviour on CREMA-D. *Recovery* is the share of clips whose plurality perceived label equals the intended one, over the 6798 clips with an unambiguous plurality;  $\alpha$  is the mean share of votes going to the winning category,  $H$  the vote entropy normalised to  $[0, 1]$ .

| Intended | $n$  | Recovery (%) | $\alpha$ | $H$  |
|----------|------|--------------|----------|------|
| Anger    | 1271 | 66.4         | 0.65     | 0.43 |
| Disgust  | 1271 | 30.2         | 0.56     | 0.55 |
| Fear     | 1271 | 35.5         | 0.62     | 0.49 |
| Happy    | 1271 | 28.9         | 0.62     | 0.48 |
| Neutral  | 1087 | 97.6         | 0.77     | 0.33 |
| Sad      | 1271 | 18.2         | 0.62     | 0.48 |
| All      | 7442 | <b>45.6</b>  | 0.64     | 0.46 |
| Chance   | —    | 16.7         | —        | —    |

### 3. THE LABEL GAP

#### 3.1. Data and label sets

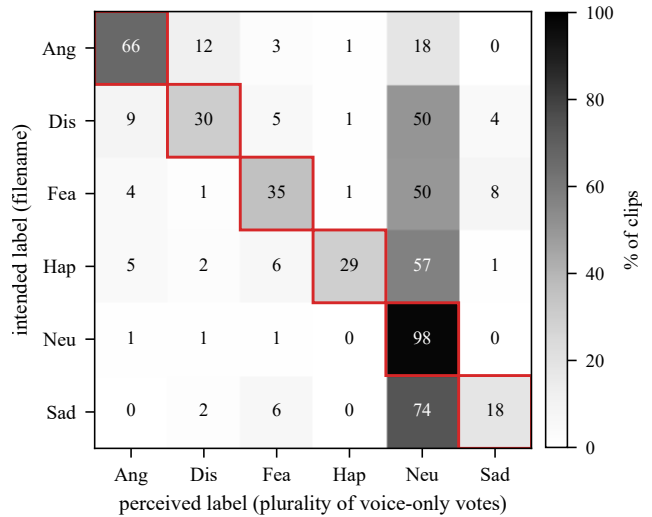
CREMA-D was rated by 2,443 crowd workers, each clip being judged in an audio condition, a visual condition and an audio-visual condition. We use the audio condition throughout. The shipped aggregation (`processedResults/tabulatedVotes.csv`) records, for every clip, how many voice-only raters chose each of the six categories; there are 68,568 such judgements, a mean of 9.2 per clip. The raw response file (`finishedResponses.csv`) carries 4,686 further voice-only ratings that the released aggregation drops; recomputing everything from the raw file gives 45.3% for the headline figure below and 18.2% for sad, so nothing here depends on which of the two files one reads.

From these we define three labels per clip. The *intended* label  $y^I$  is the filename category. The *perceived* label  $y^P$  is the plurality of the voice-only votes. The *soft* label  $y^S$  is the normalised vote distribution over the six categories. Ties for the plurality occur on 8.7% of clips; where a hard perceived label is required we break them towards the co-winning category with the higher mean rated intensity, and Section 3.2 reports the headline figure under every alternative convention we could think of, none of which changes the conclusion.

#### 3.2. How often is intent recovered?

Table 1 gives the answer per category. Over the 6798 clips with an unambiguous plurality, the perceived label equals the intended label for 45.6% of clips. Anger is recovered reasonably well and neutral almost always; the four remaining categories sit at or below 36%, and sad is at 18.2%, one and a half points above what a six-sided die would achieve. Tie handling moves the overall figure between 41.6% (every tie counted as a miss) and 47.9% (every tie containing the intended category counted as a hit), so that choice is immaterial to the argument.

Two secondary observations are worth recording because they rule out easy explanations. The gap is not an artefact of raters ignoring the task: agreement among them is well above chance, with Fleiss’  $\kappa = 0.28$  [13] and Krippendorff’s  $\alpha = 0.28$  [14]; modest, but the raters are clearly hearing something in common. Nor is it a property of the corpus as a whole. Running the same computation on the visual and audio-visual conditions gives 70.7% and 76.5% recovery respectively. The actors were, by and large, doing their job, and a viewer can tell what they were doing. What fails is the recovery of the instruction by ear—and the ear is the only channel SER has.



**Fig. 1.** Intended category (rows) against the plurality voice-only perceived category (columns), row-normalised, over the 6798 clips with an unambiguous plurality. The diagonal, which is what published CREMA-D accuracies are scored against, carries less than half the mass.

Intensity behaves as one would hope, which is a useful sanity check on the ratings themselves: for the subset of sentences recorded at three directed levels, recovery rises monotonically from 18.4% at low intensity to 55.5% at high intensity.

#### 3.3. Where the labels go

Figure 1 shows that the errors are not diffuse. Neutral is an attractor: 49.6% of clips carrying a non-neutral instruction are heard as neutral, and the perceived label is neutral for 56.6% of the corpus against 14.6% of the intended labels. Sad is the extreme case, with 71.8% of sad-intended clips heard as neutral against 18.2% heard as sad. The natural reading is that acted sadness here is realised mostly as lowered arousal, and lowered arousal in a short read sentence is hard for a listener to separate from an unemotional one. Whatever the mechanism, a system that labels every sad-intended clip *neutral* is not making a mistake by the listeners’ standard; it is agreeing with them.

#### 3.4. A human ceiling

How hard is each label to predict? For a human listener this can be answered without any model. For each of the 68,568 voice-only judgements we removed that rating, recomputed the plurality of the remaining raters for the clip, and asked whether the held-out rater matched it. A listener agrees with the actor’s instruction 40.2% of the time and with their peers 58.2% of the time. From the same audio, judged by the same people, the perceptual target is eighteen points more predictable than the intended one.

It would be natural to conclude that the intended label is simply the noisier target and that a recogniser trained on it is chasing information the recording does not contain. Section 5 shows that this conclusion is wrong, and the way in which it is wrong is the more interesting half of the story.

## 4. EXPERIMENTAL SETUP

**Representations.** We use three, chosen to span capacity rather than to be competitive. MFCC: 13 MFCC means and standard deviations (26-d). FUNC: a GeMAPS-style functional set [15] of 84 prosodic, spectral and cepstral descriptors, each summarised by eight statistics—mean, standard deviation, median, interquartile range, the 1st and 99th percentiles and their range, and a linear slope over time—together with voiced fraction, duration and jitter and shimmer proxies (677-d). SSL: frozen WavLM Base+ [16] hidden states mean-pooled over time and averaged over transformer layers 1–12 (768-d). The layer average is fixed a priori so that no part of feature extraction is tuned against either label set.

**Systems.** We pair those representations with logistic regression, a Nyström-approximated RBF classifier, histogram gradient boosting and a two-layer MLP, giving eight systems in all. They are deliberately ordinary. The aim is not to advance the state of the art on CREMA-D but to obtain the kind of spread of systems a benchmark table contains.

**Training.** Every target is a distribution over the six categories, so the three label regimes differ only in that distribution: one-hot on  $y^I$ , one-hot on  $y^P$ , or the full  $y^S$ . All systems minimise cross-entropy against it, which for a distributional target is obtained exactly by replicating a clip once per category with that category’s probability mass as its sample weight. This keeps the comparison across regimes free of any confound from the objective. Training is class-balanced by inverse frequency in every regime. This is close to a no-op for  $y^I$ , whose classes are near-uniform by construction (1,271 clips each, except neutral at 1,087), and it is essential for  $y^P$  and  $y^S$ , of which 56.6% is neutral: unbalanced training against a neutral-dominated target simply collapses onto neutral. Since the metric we report is deliberately prior-insensitive, so is the training objective, and no regime is penalised for the shape of its label distribution rather than its content.

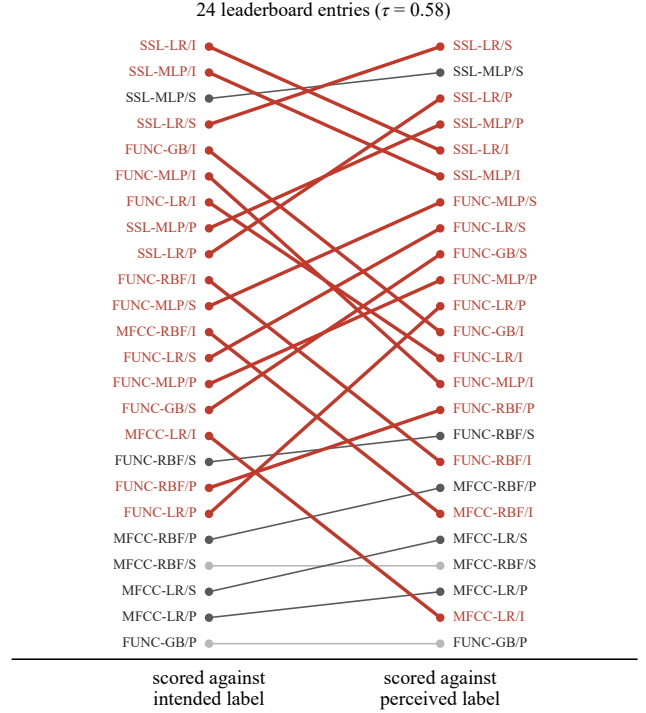
**Protocol.** Speaker-independent five-fold cross-validation over the 91 actors; no actor appears in both a training and a test fold, and within each training fold a further group of actors is held out for early stopping. Hyperparameters were fixed once and shared across regimes. We report unweighted average recall (UAR), the usual choice for class-imbalanced SER, and additionally  $R$ , the probability that a system’s output matches a randomly drawn listener vote, which is the natural criterion when the reference is a distribution.  $R$  has a human reference: two listeners rating the same clip agree 46.5% of the time. Confidence intervals come from a clip-level bootstrap with 2,000 resamples, shared across systems so that paired comparisons use common draws.

## 5. RESULTS

### 5.1. The numbers move

Table 2 gives every system under every regime. Under the standard protocol—train on the filename label, score against it—our strongest system, SSL-LR, reports 73.5% accuracy, which is the range CREMA-D papers usually occupy; the spread below it is a realistic one and not a set of straw men. Scoring those identical predictions against the perceptual label instead gives 43.8%. Across the eight systems that substitution is worth up to 29.7 points of accuracy and 14.8 points of UAR, with nothing about the model changed.

Training on the perceptual labels moves things the other way. Averaged over systems, fitting  $y^P$  rather than  $y^I$  buys 1.0 points of  $UAR_P$ , and fitting the full vote distribution  $y^S$  buys 3.4. That the



**Fig. 2.** All 24 entries—every system under every training label—ranked by UAR against the intended label (left) and against the perceived label (right). Only the answer key differs. Entries that move three or more places are drawn with heavier red lines.

soft target beats the hard perceptual one is consistent with the soft-label literature [7], and is the one place in this paper where we reproduce a known result rather than report a new one. The perceptual regimes also clear the human bar on  $R$ : the best perceptually trained entry matches a randomly drawn listener 51.6% of the time, where two listeners rating the same clip agree only 46.5% of the time. Predicting a listener is easier than being one, which is what a model fitted to the vote distribution is for.

None of this is surprising once stated—models fit the labels they are given and are flattered by being measured against them—but it establishes the size of the effect. Two papers reporting CREMA-D accuracy under the two conventions are not describing the same quantity, and nothing in a typical experimental section would reveal which convention was used.

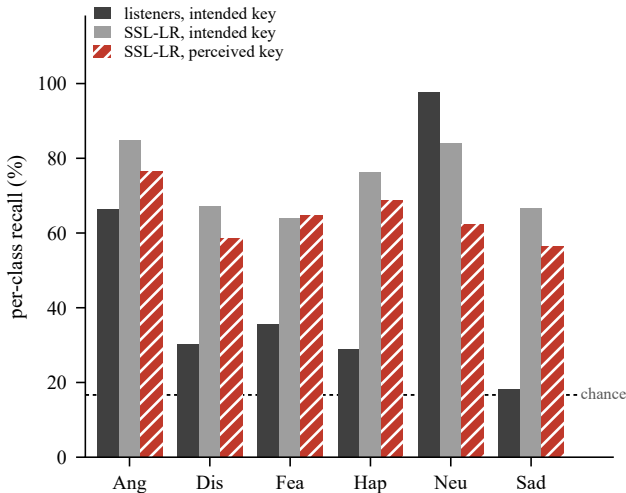
### 5.2. When the ranking moves

Start with the narrowest version of the question. Fix the training label at  $y^I$ , the standard protocol, and change only what the predictions are scored against. Here the ordering essentially survives: Kendall’s  $\tau = 0.93$ , with 1 of 28 pairs changing places, and the two systems in that pair separated by less than a tenth of a point under either key. No inversion under any single training regime survives the paired bootstrap. Architectures trained the same way remain comparable: the two keys disagree about how good a system is, not about which is better.

A benchmark table is not in that position, and Figure 2 shows why. Such a table collects results from papers that made different and largely undocumented choices about labels, so the unit that gets ranked is the entry, not the architecture. Pooling our 24 entries,  $\tau$  falls to 0.58 and 58 of 276 pairs invert. The entry that wins under

**Table 2.** Unweighted average recall (%) under speaker-independent five-fold cross-validation over the 91 actors. Each block fixes the label a system was *trained* on; within a block,  $UAR_I$  and  $UAR_P$  score the same predictions against the intended and the perceived label respectively, and  $R$  is the probability that the prediction matches one randomly drawn listener vote. Best in each column in bold.

| System   | trained on intended |             |             | trained on perceived |             |             | trained on soft |             |             |
|----------|---------------------|-------------|-------------|----------------------|-------------|-------------|-----------------|-------------|-------------|
|          | $UAR_I$             | $UAR_P$     | $R$         | $UAR_I$              | $UAR_P$     | $R$         | $UAR_I$         | $UAR_P$     | $R$         |
| FUNC-GB  | 55.8                | 49.3        | 34.8        | 34.1                 | 39.3        | <b>51.6</b> | 45.4            | 51.1        | 43.3        |
| FUNC-LR  | 54.6                | 48.2        | 33.8        | 43.7                 | 49.3        | 42.1        | 47.8            | 52.0        | 38.2        |
| FUNC-MLP | 54.6                | 48.1        | 34.6        | 45.5                 | 50.6        | 40.6        | 49.1            | 52.5        | 36.9        |
| FUNC-RBF | 51.3                | 46.2        | 35.8        | 44.3                 | 47.5        | 38.1        | 44.6            | 46.4        | 34.6        |
| MFCC-LR  | 45.2                | 42.7        | 31.4        | 39.8                 | 43.4        | 42.0        | 43.9            | 43.9        | 32.6        |
| MFCC-RBF | 48.1                | 44.1        | 33.6        | 42.1                 | 44.9        | 35.3        | 42.1            | 43.5        | 32.6        |
| SSL-LR   | <b>73.8</b>         | <b>58.9</b> | 39.2        | 53.1                 | <b>64.5</b> | 51.3        | 60.1            | <b>66.7</b> | <b>46.1</b> |
| SSL-MLP  | 72.5                | 58.1        | <b>39.5</b> | <b>53.9</b>          | 64.4        | 50.0        | <b>61.3</b>     | 66.6        | 45.3        |



**Fig. 3.** Per-class recall for the strongest system under each key, against the listeners’ own recovery of the intended label. On the categories listeners find hardest—sadness above all—the classifier reads the actor’s instruction out of the audio far better than they do.

intended labels (SSL-LR/intended) places 5th under perceived ones, and the entry that wins under perceived labels (SSL-LR/soft) places fourth under intended. Two groups reporting CREMA-D under different conventions would disagree about which system is best, and each would be right about its own.

Were the two labels separated by a constant—were perceived labels simply intended labels plus noise—every system would lose the same amount and the ordering would be untouched. It is not, because the keys reward different behaviour. A system that has learned the neutral attractor scores well against perception and poorly against intent; a system that has learned to separate the acted categories does the reverse. Model selection on CREMA-D therefore inherits whichever choice of key was made, usually silently.

### 5.3. What the two keys are asking for

We do not claim the perceptual label is correct and the intended one wrong. They set different tasks, and Figure 3 shows how differently.

Scored against intent, our strongest system recovers acted sadness for 66.6% of clips. Listeners manage 18.2%. The same pattern holds for disgust (67.2% against 30.2%) and happiness (76.2% against 28.9%): on 5 of the six categories a plain classifier reads the actor’s instruction out of the audio better than the listeners did, by as much as 48 points. It trails them only on neutral (84.1% against 97.6%), for the unremarkable reason that a class-balanced classifier

will not put half the corpus into one category. What it does not do is find a different set of categories difficult: its per-class recall correlates at  $r = 0.81$  with the listeners’ across the six. The model finds the same things hard, and is simply less troubled by them.

The acoustic trace of acted sadness, then, is present and consistent enough to be learned. What listeners fail to do is not detect it but *call it sadness*: their criterion for the label demands more than these recordings supply, and everything short of it collects in neutral. This reframes the choice of key rather than settling it. Training against the filename label is not fitting noise; it is training a system to recover the direction an actor was given through a performance that listeners hear as unemotional. That is a coherent objective and a classifier is good at it. It is not, however, the objective that the phrase *speech emotion recognition* is normally taken to name, and a number obtained under it should not be read as a claim about what anyone hears.

## 6. DISCUSSION

The narrow recommendation is that CREMA-D results should state which label set they use, and that the perceptual labels—which require no new annotation, only reading a file that ships with the corpus—should be reported alongside the filename labels. Doing so costs nothing and makes results comparable.

The broader point concerns acted corpora generally. Such a corpus carries two things at once: the instruction an actor was given, and the impression the performance leaves. Both turn out to be recoverable from the audio, which is what makes the choice between them a real one rather than a choice between signal and noise. A corpus that ships perceptual ratings makes that choice explicit; CREMA-D ships them, and the community has settled by default on the key that puts the reference label out of reach of the listeners the system is nominally imitating.

**Limitations.** The perceptual labels are themselves a consensus over roughly nine raters, with the reliability reported in Section 3.2; a plurality over nine crowd workers is a noisy estimate of a distribution, and clips near a tie are close to arbitrary. Our systems are conventional and CPU-scale, so the reordering is demonstrated on the kind of spread found in benchmark tables rather than among frontier models; whether it also occurs among a set of near-identical large models is open. Finally, everything here concerns one corpus. CREMA-D is the convenient case because it distributes both labels for the same clips, but the question it raises—which reading of an acted corpus is being scored—applies wherever directed emotion is used as ground truth.

## Compliance with ethical standards

This is a secondary analysis of CREMA-D [1], a corpus released publicly by its authors under the Open Database License. No new recordings and no new human ratings were collected for this work, and no identifying information about the actors or the raters is used beyond the demographic fields already distributed with the corpus.

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